

A counterexample to Proposition 2.11 of ‘Classicism’

Fitelson’s model, as an intensional action model

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reworking a construction by Branden Fitelson, with Claude and GPT-6 Astra

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Abstract

Proposition 2.11 of ‘Classicism’ says that $\Box$ Atomicity, Boolean Completeness and BF jointly imply Rigid Comprehension. Branden Fitelson has sent a model, produced with Claude and GPT-6 Astra, equally, and checked in Isabelle/HOL, in which $\Box$ Atomicity, $\Box$ Boolean Completeness, $\Box$ BF and $\Box$ Actuality all hold but Rigid Comprehension fails for properties of propositions. This note presents the same model as an intensional action model in the sense of BCsansRC (§2), where it becomes quite simple: there is one individual, the root sees each other world by exactly one arrow, and the only non-trivial constraint is that at the root, what a property does at one particular world u is the ultralimit of what it does at countably many other worlds. §§1–2 set out the model and what each part of it is for; §3 verifies the positive principles; §4 refutes Rigid Comprehension; §5 runs the proof in footnote 42 inside the model to show where it breaks.

1 The idea

Recall from BCsansRC (Section 2 below gives the details) that in an intensional action model the Boolean operations on each relational type are set operations, *provided* the domains are closed under them. Finite Boolean combinations are always available, because they are denotations of terms. Infinite intersections and unions need not be. Boolean Completeness asks only that every family have a greatest lower bound *in the domain*, and if the domain is not closed under intersection that bound can be strictly smaller than the intersection. Then the transport of the bound to a later world can differ from the bound of the transported family.

The proof of Proposition 2.11 takes the least upper bound $\mathbf{X}^*$ of the haecceities of the X s as its candidate rigid property. $\mathbf{X}^*$ is coextensive with X and *weakly* rigid, i.e. it satisfies the matrix of Rigid at the actual world (this much follows from Boolean Completeness and Actuality alone: BCsansRC, Proposition 1). Rigidity requires that matrix at every world, so what is needed is that $\mathbf{X}^*$ *remain* a least upper bound of the relevant haecceities when transported to other worlds. The model makes that fail, as follows.

- There is a world u that the actual world o sees only ‘in the limit’: at o , the truth value of any proposition at u is the $\mathcal{U}$ -limit of its truth values at worlds $1, 2, 3, \dots$, for a non-principal ultrafilter $\mathcal{U}$. So infinite meets and joins at o are computed coordinatewise everywhere except at u , where they are recomputed as limits.
- There is a world s that sees u but none of the worlds n . From s , u is an ordinary world, and what happens there is not constrained by anything.
- At o , the least upper bound of a family of haecceities can pick up, at u , a value that no member of the family has there. The members can be jointly spread over the worlds n in a way that has limit ‘true’ at u , while each member is ‘false’ at u . From o the extra value makes no difference: every persistent property containing the members has it too, so the least upper bound is still least. From s , which sees u directly and none of the worlds n , it is exactly what separates the least upper bound from the haecceities it bounds.

Individuals play no role (there is just one), and the counterexample is at type $t \rightarrow t$. All the worlds other than o are finite, and the worlds n and u all carry the same finite full type structure, so the ultralimits are taken in fixed finite sets. The only non-constructive ingredient is the ultrafilter: the intensional presentation, unlike the functional one in Fitelson’s note, needs no further use of choice.

2 The model

2.1 Intensional action models

We use the definitions of BCsansRC §2. In brief: a rooted category $\langle \mathcal{C}, W_0 \rangle$; for each object W and relational type $\tau = \sigma_1 \rightarrow \dots \rightarrow \sigma_k \rightarrow t$, the *full* intension space $W^{[\tau]}$ is the set of all sets of tuples $\langle \mathbf{x}_1, \dots, \mathbf{x}_k, h \rangle$ with $h : W \rightarrow V$ and $\mathbf{x}_i \in V^{\sigma_i}$; the domain W^τ is a subset of it, and an arrow $h : W \rightarrow V$ transports $\mathbf{A} \in W^\tau$ to $h^\tau \mathbf{A} = \{ \langle \vec{x}, i \rangle \mid \langle \vec{x}, i \circ h \rangle \in \mathbf{A} \} \in V^\tau$. $\text{ext } \mathbf{A} = \{ \vec{x} \mid \langle \vec{x}, \mathbf{1}_W \rangle \in \mathbf{A} \}$. A premodel is a model when every term denotes, at every arrow out of W_0 , an element of the

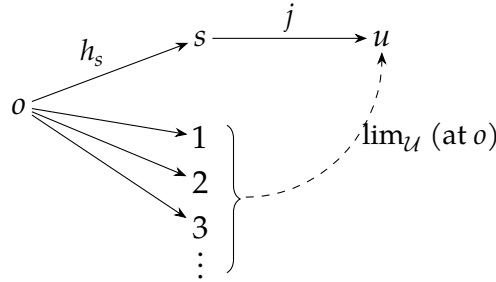
domain there; Classicism is sound for these models. We use three facts from BCsansRC:

- (a) (Booleans are set operations; Lemma 9.) In a model, $\wedge_\tau, \vee_\tau, \neg_\tau$ denote intersection, union and complement, and $\mathbf{A} \leq_\tau \mathbf{B}$ is true iff $\mathbf{A} \subseteq \mathbf{B}$.
- (b) (Transport; Lemma 11.) $\llbracket A \rrbracket_{i \circ h, i \circ g} = i^\sigma \llbracket A \rrbracket_{h, g}$. The proof is by induction on terms and uses only the existence of the denotations involved.
- (c) (Rigidity criterion; Lemma 14.) $\mathbf{A} \in W_0^\tau$ is rigid iff $\mathbf{A}$ is *persistent*, i.e. transport-closed ($\langle \vec{x}, h \rangle \in \mathbf{A}$ implies $\langle i\vec{x}, i \circ h \rangle \in \mathbf{A}$), and for every arrow w out of W_0 and every persistent $\mathbf{B} \in (\text{trg } w)^\tau$, $\text{ext}(w^\tau \mathbf{A}) \subseteq \text{ext } \mathbf{B}$ implies $w^\tau \mathbf{A} \subseteq \mathbf{B}$.

2.2 The category

The objects are o (the root), s, u , and n for each $n \in \mathbb{N}_{>0}$. Besides identities, the arrows are

$$h_s: o \rightarrow s, \quad j: s \rightarrow u, \quad h_u := j \circ h_s: o \rightarrow u, \quad h_n: o \rightarrow n.$$



So o has exactly one arrow to each object. We may therefore identify the worlds of the model (the arrows out of o) with the objects, writing $h_o := \mathbf{1}_o$. From s the worlds are s and u ; u and each n see only themselves.

2.3 Extensions at worlds

By BCsansRC Lemma 5(iii), an element $\mathbf{A} \in o^{[\tau]}$ is determined by the extensions of its transports, one for each world. Write

$$\mathbf{A}_v := \text{ext}(h_v^{[\tau]} \mathbf{A}) = \{\vec{x} \mid \langle \vec{x}, h_v \rangle \in \mathbf{A}\} \quad (v \text{ an object})$$

for the *extension of $\mathbf{A}$ at v* . In general one speaks of the extension of $\mathbf{A}$ at an arrow h , namely $\text{ext}(h^{[\tau]} \mathbf{A})$; here each world is reached by exactly one arrow, so

we can index by worlds. In particular $\mathbf{A}_o = \text{ext } \mathbf{A}$. An element of $o^{[\tau]}$ is then just an arbitrary choice of an extension $\mathbf{A}_v \subseteq \prod_i v^{\sigma_i}$ at each world v . Likewise an element $\mathbf{B} \in s^{[\tau]}$ is a pair of extensions, $\mathbf{B}_s$ at s and $\mathbf{B}_u = \text{ext}(j^{[\tau]}\mathbf{B})$ at u . Transport keeps the extensions at the worlds still reachable and drops the rest: $(h_s^{[\tau]}\mathbf{A})_s = \mathbf{A}_s$ and $(h_s^{[\tau]}\mathbf{A})_u = \mathbf{A}_u$, and the transport of $\mathbf{A}$ to a terminal world v is the intension whose extension is $\mathbf{A}_v$.

2.4 The domains

Let M be the full simple type structure over one individual: $M_e = \{*\}$ and $M_{\sigma_1 \rightarrow \dots \rightarrow \sigma_k \rightarrow t} = \mathcal{P}(M_{\sigma_1} \times \dots \times M_{\sigma_k})$. Each M_σ is finite. In particular $M_t = \{\emptyset, \{\langle \rangle\}\}$, which we write $\{0, 1\}$.

- *Individuals.* $v^e = \{*\}$ at every object v .
- *The terminal worlds u and n : full.* $v^\tau := v^{[\tau]}$. Since the only arrow out of v is 1_v , an element is determined by its extension, and $\mathbf{A} \mapsto \text{ext } \mathbf{A}$ (applied recursively to the arguments) identifies v^σ with M_σ . We use this identification throughout, so that the extensions of a root element at u and at every n all lie in the same finite set. At u and n we write $\mathbf{t}$ and $\mathbf{f}$ for the true and false propositions, i.e. 1 and 0.
- *The world s : full.* $s^\tau := s^{[\tau]}$. For example, s^t has four elements, one for each pair of truth values at s and at u . We write $\mathbf{f}_s$ for the proposition false at both.
- *The root: full except at u .* For relational τ , by recursion on τ ,

$$o^\tau := \{\mathbf{A} \in o^{[\tau]} \mid \mathbf{A}_u = \lim_{\mathcal{U}} \mathbf{A}_n\},$$

where for a sequence (a_n) in a finite set, $\lim_{\mathcal{U}} a_n$ is the unique a with $\{n \mid a_n = a\} \in \mathcal{U}$.

So a root element is free at o , at s and at each n , while its extension at u is dictated by the limit of its extensions at the n s. For propositions: a root proposition $\mathbf{p}$ is a choice of truth values $\mathbf{p}_o, \mathbf{p}_s, \mathbf{p}_1, \mathbf{p}_2, \dots$ arbitrarily, with $\mathbf{p}_u = 1$ iff $\{n \mid \mathbf{p}_n = 1\} \in \mathcal{U}$. The transports of root elements land in the full domains of the other objects, and j transports from one full domain to another, so this is a premodel.

The arrangement does three jobs:

- The limit constraint is *closed under every finite operation* (Lemma 1), so every term denotes an element of the domain (Lemma 2). It is *not* closed under infinite intersections and unions, which is the whole point.

- The constraint is imposed *only at the root*. At s the world u is unconstrained, so a property evaluated from s can treat u independently of the worlds n , which s cannot see.
- The worlds n give the root's elements room to differ from one another at infinitely many independent places, while all agreeing at u .

Lemma 1 (Limits commute with finite operations). If $A_1, \dots, A_k, B$ are finite and $f: A_1 \times \dots \times A_k \rightarrow B$, then $\lim_{\mathcal{U}} f(a_n^1, \dots, a_n^k) = f(\lim_{\mathcal{U}} a_n^1, \dots, \lim_{\mathcal{U}} a_n^k)$. Changing finitely many terms of a sequence does not change its limit.

Proof. Each a_n^i equals $\lim_{\mathcal{U}} a_n^i$ on a set in $\mathcal{U}$; on the intersection of these k sets the two sides agree. $\mathcal{U}$ contains every cofinite set. $\square$

Lemma 2. $\mathfrak{M}$ is an intensional action model. Hence Classicism holds in it.

Proof. We show by induction on terms A that $\llbracket A \rrbracket_{h,g}$ exists and belongs to $(\text{trg } h)^\sigma$ for every arrow h out of o and every adequate assignment g . Existence comes from the inductive hypothesis, as in any premodel. If $\text{trg } h \neq o$ the domain is full, so membership is automatic. It remains to treat $h = \mathbf{1}_o$, with A of relational type τ . By (b),

$$\llbracket A \rrbracket_{\mathbf{1}_{g_o}} = \text{ext} \llbracket A \rrbracket_{h_v, h_v \circ g}.$$

At a terminal world v every arrow is the identity, so the clauses of the interpretation function reduce to the Tarskian clauses for the full type structure M . Hence $\llbracket A \rrbracket_{\mathbf{1}_{g_v}}$ is the value of A in M when each free variable and constant of A is assigned the extension at v of its value at the root. A has finitely many free variables and constants. Their root values lie in root domains, so each has extension at u equal to the limit of its extensions at the ns . By Lemma 1 there is a set in $\mathcal{U}$ on which all of them have the same extension at n as at u . For n in that set, A gets the same value in M at n as at u . So $\llbracket A \rrbracket_{\mathbf{1}_{g_u}} = \lim_{\mathcal{U}} \llbracket A \rrbracket_{\mathbf{1}_{g_n}}$. $\square$

3 What holds

Lemma 3 (The root algebras). For relational τ , $\mathbf{A} \mapsto (\mathbf{A}_o, \mathbf{A}_s, (\mathbf{A}_n)_{n \geq 1})$ is an isomorphism of partial orders from $(o^\tau, \subseteq)$ onto the product of the powersets of $\prod_i o^{\sigma_i}$, of $\prod_i s^{\sigma_i}$, and (for each n) of $\prod_i M_{\sigma_i}$. Consequently:

- (i) every family $\mathcal{A} \subseteq o^\tau$ has a meet and a join in o^τ . They are computed by intersection and union at o , s and each n , and at u by taking the limit of the results at the ns . Finite meets and joins are intersections and unions; infinite ones need not be (Remark 5 gives an example);

- (ii) the atoms of o^τ are the elements with exactly one tuple in exactly one of their extensions at o, s or some n (and hence empty extensions elsewhere, including at u). Every nonzero element is above one.

Proof. The map is injective because the extension at u is determined by the others, and surjective because any choice of the extensions at o, s and the ns , completed by their limit at u , is a root element. It preserves and reflects inclusion because $\lim_{\mathcal{U}}$ is monotone: if $\mathbf{A}_n \subseteq \mathbf{B}_n$ for all n then $\lim_{\mathcal{U}} \mathbf{A}_n \subseteq \lim_{\mathcal{U}} \mathbf{B}_n$. (i) and (ii) are then facts about products of powerset algebras. Finite meets and joins are intersections and unions by Lemma 1. For (ii), note that a nonzero element must have a nonempty extension at o, s or some n , since if all of these are empty so is the limit. $\square$

At s, u and each n the domains are finite powerset algebras. So, using (a):

Proposition 4. At every world of $\mathfrak{M}$ and every relational type, Boolean Completeness, Atomicity and Actuality hold. BF holds at every world and every type. Hence $\square$ Boolean Completeness, $\square$ Atomicity, $\square$ Actuality and $\square$ BF hold in $\mathfrak{M}$.

Proof. Boolean Completeness and Atomicity. At a world v , ‘ y is a GLB of X ’ says (by (a)) that y is the greatest element of v^τ , under $\subseteq$, below every element of $\text{ext } X$. The domains are complete lattices under $\subseteq$ (Lemma 3(i) at o ; powersets elsewhere), so such a y exists. Atomicity is Lemma 3(ii), and is trivial at the finite worlds.

Actuality. The root proposition true at o and false everywhere else (so 0 at u) is a true atom, contained in every true proposition. At s the corresponding proposition is the one true at s only; the terminal worlds are trivial.

BF. Every arrow acts surjectively on every type. To lift $\mathbf{B} \in s^\tau$ along h_s , take the root element with extensions $\mathbf{B}_s$ at s , $\mathbf{B}_u$ at u and at every n , and anything at o . To lift $\mathbf{a} \in M_\tau$ along h_n , take the root element with extension $\mathbf{a}$ at n and some fixed $\mathbf{c}$ at every other m and at u ; the limit is unaffected by the one exception. h_u and j are similar. Surjectivity gives BF in the usual way (‘Classicism’, Proposition 3.24(ii)). If $\forall x \square Px$ holds at h and i is an arrow out of $\text{trg } h$, every $\mathbf{a}$ in the domain at $\text{trg } i$ is ix for some x at $\text{trg } h$. $\square Px$ then gives Pa at $i \circ h$, so $\forall x Px$ holds there. $\square$

Remark 5 (Transport is not complete). $h_s^\tau: o^\tau \rightarrow s^\tau$ preserves finite meets and joins (Lemma 3(i)) but not infinite ones. Let $\mathbf{p}^{(k)}$ be the root proposition true exactly at the worlds $n \geq k$ (and so true at u). Each $h_s \mathbf{p}^{(k)}$ is the proposition at s true only at u . But $\bigwedge_k \mathbf{p}^{(k)} = \perp$, since the intersection of their extensions at each n is empty, so $h_s(\bigwedge_k \mathbf{p}^{(k)}) = \perp \neq \bigwedge_k h_s \mathbf{p}^{(k)}$. From o , the world u is accessible only through the limit; from s it is directly accessible. This is what the proof of Proposition 2.11 cannot cope with.

4 Rigid Comprehension fails

Let $\mathbf{f}_s \in s^t$ be the proposition false at s and at u , and

$$A_0 := \{\mathbf{p} \in o^t \mid \mathbf{p}_s = \mathbf{p}_u = 0\} = \{\mathbf{p} \mid \mathbf{p}_s = 0 \text{ and } \{n \mid \mathbf{p}_n = 1\} \notin \mathcal{U}\}.$$

These are the root propositions false at s and at u ; each is $\mathbf{f}_s$ at s . But at any *single* world n , a member of A_0 can have either truth value: the proposition true only at n is in A_0 , since one world does not change the limit. Let $\mathbf{X} \in o^{t \rightarrow t}$ have extension A_0 at o and empty extensions elsewhere. (It is in $o^{t \rightarrow t}$, since its extension at u is the limit of its empty extensions at the ns .)

Theorem 6. No rigid $\mathbf{Z} \in o^{t \rightarrow t}$ is coextensive with $\mathbf{X}$. So Rigid Comprehension at type $t \rightarrow t$ fails in $\mathfrak{M}$.

Two properties do the work. The first is a root property:

$$\mathbf{X}^* : \quad \mathbf{X}_o^* = A_0, \quad \mathbf{X}_s^* = \{\mathbf{f}_s\}, \quad \mathbf{X}_n^* = \{0, 1\} \text{ (all } n), \quad \mathbf{X}_u^* = \{0, 1\}.$$

It is in $o^{t \rightarrow t}$, and it is persistent: the transport of a member of A_0 is $\mathbf{f}_s$ at s , something at each n , and 0 at u , all of which are in $\mathbf{X}^*$'s extensions at those worlds; and j sends $\mathbf{f}_s$ to 0. The second is the haecceity of $\mathbf{f}_s$ at s :

$$\mathbf{Y} \in s^{t \rightarrow t} : \quad \mathbf{Y}_s = \{\mathbf{f}_s\}, \quad \mathbf{Y}_u = \{0\},$$

which is also persistent.

Proof of Theorem 6. Suppose $\mathbf{Z}$ is rigid with $\mathbf{Z}_o = \text{ext } \mathbf{Z} = A_0$.

- (1) *Persistence at the worlds n .* $\mathbf{Z}$ is transport-closed, so for each $\mathbf{p} \in A_0$ it contains the transports of $\mathbf{p}$: $\mathbf{f}_s \in \mathbf{Z}_s$, $\mathbf{p}_n \in \mathbf{Z}_n$ for every n , and $0 \in \mathbf{Z}_u$. Since members of A_0 take both values at each n , $\mathbf{Z}_n = \{0, 1\}$ for every n .
- (2) *The limit.* $\mathbf{Z}$ is a root element, so $\mathbf{Z}_u = \lim_{\mathcal{U}} \mathbf{Z}_n = \{0, 1\}$. In particular $1 \in \mathbf{Z}_u$, although no member of A_0 is true at u .
- (3) *Rigidity at o .* $\mathbf{X}^*$ is persistent and $\text{ext } \mathbf{Z} = A_0 = \text{ext } \mathbf{X}^*$, so by (c) with $w = \mathbf{1}_o$, $\mathbf{Z} \subseteq \mathbf{X}^*$. Hence $\mathbf{Z}_s \subseteq \{\mathbf{f}_s\}$, and so $\mathbf{Z}_s = \{\mathbf{f}_s\}$ by (1).
- (4) *Rigidity at s .* $\text{ext}(h_s \mathbf{Z}) = \mathbf{Z}_s = \{\mathbf{f}_s\} = \text{ext } \mathbf{Y}$, and $\mathbf{Y}$ is persistent. So by (c) with $w = h_s$, $h_s \mathbf{Z} \subseteq \mathbf{Y}$, and in particular $\mathbf{Z}_u \subseteq \mathbf{Y}_u = \{0\}$. This contradicts (2). $\square$

What happened. The actual instances of $\mathbf{X}$ are scattered across the worlds n : at each single n they take both truth values, but on any $\mathcal{U}$ -large set of worlds they are mostly false. A persistent property containing them must contain both truth values at every n , and the root's limit constraint then makes it contain 'true' at

u . From the root's point of view that costs nothing, because every persistent property containing the instances has the same feature. From s , however, the instances all look the same ($\mathbf{f}_s$, which is false at u), and rigidity at s demands that $\mathbf{Z}$ be included in the persistent property ($\mathbf{Y}$) that contains only what $\mathbf{f}_s$ is at u . The outer box in Rigid is essential. $\mathbf{X}^*$ is *weakly rigid*: it satisfies the matrix of Rigid without the leading $\Box$ (BCsansRC §3.1), since step (3) shows it is the least persistent property extending A_0 . But the condition fails at s , so it is not rigid.

Remark 7 ($\mathbf{X}^*$ is the least upper bound of the haecceities). The haecceity of $\mathbf{p} \in A_0$ has extensions $\{\mathbf{p}\}$ at o , $\{\mathbf{f}_s\}$ at s , $\{\mathbf{p}_n\}$ at n , and $\{0\}$ at u . By Lemma 3(i) their join has extensions A_0 , $\{\mathbf{f}_s\}$ and $\bigcup_{\mathbf{p}} \{\mathbf{p}_n\} = \{0,1\}$, and so $\lim_{\mathcal{U}} \{0,1\} = \{0,1\}$ at u , where the union would give only $\{0\}$. So $\mathbf{X}^*$ is exactly the property that the proof of Proposition 2.11 proposes as the rigid coextension of $\mathbf{X}$. It has the features BCsansRC Proposition 1 derives from Boolean Completeness and Actuality: it is coextensive with $\mathbf{X}$ and weakly rigid. The missing ingredient is the leading $\Box$ in Rigid. At s , Boolean Completeness supplies a least upper bound of the haecceities of the things that are $\mathbf{X}^*$ there. That is $\mathbf{Y}$, which is coextensive with $h_s \mathbf{X}^*$ at s and below it, but not identical to it.

5 Where the proof of Proposition 2.11 breaks

Parts (i)–(iii) of footnote 42 are correct, and here they are verified by $\mathbf{X}^*$. Part (iv) argues for inextensibility. Suppose $\Diamond(\forall x(\mathbf{X}^*x \rightarrow \Box Yx) \wedge \Diamond(\mathbf{X}^*z \wedge \neg Yz))$. Take an atom w entailing this, and an atom w' that w entails to be an atom below $\mathbf{X}^*z \wedge \neg Yz$. Then ‘without loss of generality’ take w' to be the GLB of the propositions p such that $w \leq (p = w')$. In the model:

- Y : the root property with extensions $\emptyset$ at o , $\{\mathbf{f}_s\}$ at s , $\{0\}$ at each n and at u (a root element by the limit condition);
- z : the root proposition $\mathbf{q}$ true exactly at the worlds n and hence at u ;
- w : the root atom true exactly at s . Entailment by w is truth at s ;
- w' : $\mathbf{q}$ itself. At s , $h_s \mathbf{q}$ is the atom ‘false here, true at u ’. The only thing that is $\mathbf{X}^*$ at s is $\mathbf{f}_s$, which necessarily has Y ; and $\mathbf{q}$ at u is $1 \in \mathbf{X}_u^* \searrow Y_u$. So w entails $\forall x(\mathbf{X}^*x \rightarrow \Box Yx) \wedge \text{Atom}(\mathbf{q}) \wedge \mathbf{q} \leq (\mathbf{X}^*\mathbf{q} \wedge \neg Y\mathbf{q})$.

The normalization. $w \leq (\mathbf{p} = \mathbf{q})$ iff $h_s \mathbf{p} = h_s \mathbf{q}$ iff $\mathbf{p}_s = 0$ and $\{n \mid \mathbf{p}_n = 1\} \in \mathcal{U}$. These propositions have empty intersection at o , s and every n (each n is omitted by a cofinite set), so their GLB is $\perp$. So $w'' = \perp$, and $w \not\leq (w'' = \mathbf{q})$: replacing w' by w'' loses the property of being an atom below $\mathbf{X}^*z \wedge \neg Yz$ according to w . This is

Remark 5 again. The normalization step needs h_s to preserve the meet, and it does not.

Why the step matters. It is needed for the inference ' w ' must in fact entail Yu' , which turns what w says about w' into a fact about w' at every world. Without the normalization this fails in the model. For each X -thing $\mathbf{p}$, w does entail $\mathbf{q} \leq Y\mathbf{p}$. But $\mathbf{q}$ is true at every world n , and at a world n where $\mathbf{p}_n = 1$, $Y\mathbf{p}$ is false. So $H := \lambda x.(\mathbf{q} \rightarrow Yx)$ is not an upper bound of the haecceities, and the contradiction is not reached. What w (at s) says about $\mathbf{q}$ concerns only its behaviour at u ; its behaviour at the worlds n , which s cannot see, is unconstrained by it.

6 Remarks

1. *Verification.* Fitelson reports an Isabelle/HOL proof (HOL-ZF, with the library's free ultrafilter and no added axioms) that the Rigid Comprehension instance for properties of propositions is not derivable from Classicism plus $\Box$ Atomicity, Boolean Completeness at every relational type, and BF at every type, for the model in its functional presentation: [https://github.com/fitelson/hom-isabelle/tree/4042cc0/Applications/2.11,theorem proposition_2_11_refuted](https://github.com/fitelson/hom-isabelle/tree/4042cc0/Applications/2.11,theorem%20proposition_2_11_refuted). Not formalized there: the boxed forms of Boolean Completeness and BF, and the identification of $\mathbf{X}^*$ as the least upper bound.
2. *Relation to earlier work.* The propositional part of this model (worlds o, s, u, n , with u governed by an ultrafilter) appears in the Lean-Classicism verification notes of 28 September, as a structure in which the normalization step of footnote 42 fails. It was left open there whether it extends to a model of Classicism with all higher types. The present model is that extension.
3. *A citation.* Fitelson's note also points out that footnote 42(ii) should cite Proposition 2.7 rather than 2.6, which presupposes C5. The conclusion of that step is correct all the same: Atomicity (at type t) and Boolean Completeness imply Actuality, and indeed Atomicity and Weak Inextensible Comprehension already do (Logical Maps, Classicism, result `atomicity-t-and-weakly-inextensible-comprehension-imply-actuality`).